

TP 2 : Langage de Manipulation de Données 1 (LMD):
NERMINE KIROUANE ISIL B GROUP 1:

1. Créez une base de données nommée **USTHB_CS_DB**. Connectez-vous à cette base

```
mysql> CREATE DATABASE USTHB_INFO_DB;  
Query OK, 1 row affected (0.01 sec)
```

I used the commande **create database (name of the base);**

2. Exécutez le fichier **USTHB_INFO.sql** fourni. Affichez la liste des tables.

```
mysql> connect usthb_info_db;  
Connection id: 15  
Current database: usthb_info_db  
  
mysql> source C:/Users/HP/Downloads/USTHB_INFO.sql  
Query OK, 0 rows affected (0.04 sec)  
  
Query OK, 0 rows affected (0.15 sec)  
  
Query OK, 0 rows affected (0.15 sec)
```

After connectiong to the base using the comande **connect usthb_info_db;**

i used the **commande** source to to execute the script sql script

3. Affichez la structure de la table **Section**. Ajoutez la contrainte de clé primaire pour cette table

```
mysql> describe section;
+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| year_study     | varchar(50)   | YES  |     | NULL     |       |
| specialty_code | varchar(20)   | YES  | MUL | NULL     |       |
| section_name   | varchar(5)    | YES  |     | NULL     |       |
+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

i used the commande **describe section;**

```
3 rows in set (0.01 sec)

mysql> alter table section
-> add constraint id_full_sec
-> primary key (year_study, specialty_code,
section_name);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

i used here the commande to make a big id that have multiple attribut

i used:

alter table section

-> add constraint id_full_sec

-> primary key (year_study, specialty_code,

4. La table Field existe dans la base de données mais son nom n'est pas correct. Renommez cette table en Specialty et créez les tables manquantes selon le schéma fourni;

```
3 rows in set (0.00 sec)

mysql> alter table field
-> rename specialty;
Query OK, 0 rows affected (0.03 sec)
```

i used the commande to rename the table into field i used the commande :

alter table (name table)

rename (new name);

5. Ajoutez une contrainte sur l'attribut program_type de la table Speciality afin que sa valeur soit "Licence", "Master", "Doctorat" ou "Engineering"

```
mysql> alter table speciality
-> add constraint program_types
-> check ( program_type IN ('Licence', 'Master',
'Doctorat', 'Engineering'));
```

i used the command to let only this values be as program type only those i used the commande

alter table (name table)
add constraint program_types (i name the constraint)
check (program_type-name of attribut) IN (the values chosen);

6. Insérez les tuples suivants : Tables Tuples Department ("DIASD", "Intelligence Artificielle et Science de donnees") ("2nd year", "ISIL", "B") Section Speciality ("ACAD", 'Informatique Academique', "Licence", "DSINF") Section ("3rd year", "ACAD", "C") Student (1003, "Karim Benali", 21, "3rd year", "ISIL", "B") Student (1001, "Amine Haddad", 19, "2nd year", "ISIL", "B") Student (1002, "Lina Kaci", 19, "2nd year", "ISIL", "B") Que remarquez-vous? Expliquez

```
mysql> insert into department values ('DIASD', 'Intelligence
Artificielle et Science de donnees');
Query OK, 1 row affected (0.01 sec)

mysql> insert into specialty values ( ('ACAD', 'Informatique
Academique', 'Licence', 'DSINF'));
ERROR 1064 (42000): You have an error in your SQL syntax; check the
manual that corresponds to your MySQL server version for the right
syntax to use near '' at line 1
mysql> insert into specialty values ('ACAD', 'Informatique
Academique', 'Licence', 'DSINF');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key
constraint fails (`usthb_info_db`.`specialty`, CONSTRAINT
`fk_field_dept` FOREIGN KEY (`department_ID`) REFERENCES
`department` (`department_code`))
mysql> insert into section values ('2nd year', 'ISIL', 'B') ;
ERROR 1452 (23000): Cannot add or update a child row: a foreign key
constraint fails (`usthb_info_db`.`section`, CONSTRAINT
`fk_section_spec` FOREIGN KEY (`specialty_code`) REFERENCES
`specialty` (`specialty_code`))
mysql> insert into section values ('3rd year', 'ACAD', 'C');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key
constraint fails (`usthb_info_db`.`section`, CONSTRAINT
`fk_section_spec` FOREIGN KEY (`specialty_code`) REFERENCES
`specialty` (`specialty_code`))
mysql> // i notice that insert for some types of values do not work
i guess bcuz as exmaple specialty the tuples i did insert is like
diasd and iaesdd and speciality have department id it foriegn key
```

multiple errors occurred but the first one did work

the other ones send error bcuz we need to chose as foreign keys values that do existe before
and i used the commande

**INSERT INTO (table name)
VALUES (values chosen that existe before (foriegnkeys));**

7. Insérez les tuples suivants : Department ("DSINF", "Département des Systèmes Informatiques") Speciality ("ISIL", "Ingénierie des Systèmes d'Information et Logiciels", "Licence", "DSINF")

```
mysql> insert department values ('DSINF', 'Département des Systèmes Informatiques');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> insert into specialty values ('ISIL', 'Ingénierie des Systèmes d'Information et Logiciels', 'Licence', 'DSINF');  
Query OK, 1 row affected (0.01 sec)
```

8. Réinsérez les tuples de la question 6

```
mysql> insert into specialty values ('ISIL', 'Ingénierie des Systèmes d'Information et Logiciels', 'Licence', 'DSINF');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> insert into specialty values ('ACAD', 'Informatique Academique', 'Licence', 'DSINF');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> insert into section values ('2nd year', 'ISIL', 'B') ;  
Query OK, 1 row affected (0.00 sec)  
  
mysql> insert into section values ('3nd year', 'ACAD', 'C') ;  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO STUDENT (1003, 'Karim Benali', 21, '3nd year', 'ISIL', 'B');  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1003, 'Karim Benali', 21, '3nd year', 'ISIL', 'B')' at line 1  
mysql> INSERT INTO STUDENT VALUES (1003, 'Karim Benali', 21, '3nd year', 'ISIL', 'B');  
ERROR 1146 (42S02): Table 'usthb_info.db.student' doesn't exist  
mysql> show tables;  
+-----+  
| Tables in usthb_info.db |
```

and i created to student

```
mysql> CREATE TABLE student (
->     student_ID INT PRIMARY KEY,
->     full_name VARCHAR(50),
->     age INT CHECK (age > 0),
->     year_study varchar(50),
->     specialty_code varchar(20),
->     section_name varchar(5),
-> FOREIGN KEY (year_study, specialty_code, section_name)
REFERENCES section (year_study, specialty_code, section_name));
Query OK, 0 rows affected (0.08 sec)

mysql> show tables
```

9. Modifiez le nom de la spécialité "ACAD" en "Informatique Générale".

```
mysql> update specialty
-> set specialty_name = 'Informatique Générale'
-> where specialty_code = 'ACAD';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

i did update the specialty name to using the commande

```
update specialty
set name_of_col_i_need_to_change = 'value'
where name_col = 'value'
```

10. Augmentez l'âge de tous les étudiants de 1 an.

```
mysql> update student
-> set age = 20
-> where full_name = 'Amine Haddad';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update student
-> set age = 20
-> where full_name = 'Lina Kaci';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

i did update the age with the same commandes ;

```
update specialty
set name_of_col_i_need_to_change = 'value'
where name_col = 'value'
```

11. Supprimez tous les étudiants dont l'âge est supérieur à 21 ans

i couldnt even insert it at first so it deleted bcuz there is no tuple with isil 3rd year

```
mysql> select * from section;
+-----+-----+-----+
| year_study | specialty_code | section_name |
+-----+-----+-----+
| 3rd year   | ACAD           | C            |
| 2nd year   | ISIL           | B            |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

12. Essayez de supprimer la spécialité "ISIL" de la table Speciality. Que se passe-t-il? Justifiez

```
mysql> delete from specialty
-> where specialty_code = 'ISIL';
```

i did deleted the tuple using the command:

**delete from tablename
where col = 'val'**

13. Essayez d'insérer une nouvelle spécialité de votre choix avec le program_type "Licence pro". Que se passe-t-il? Qu'est-ce qui empêche cette insertion

```
mysql> alter table specialty
-> drop check program_types;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table specialty
-> add constraint program_types
-> check ( program_type IN ('Licence', 'Licence pro', 'Master',
'Doctorat', 'Engineering'));
Query OK, 2 rows affected (0.10 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

a problem occurred so the dropped the choice and added licence pro
this commandes i didnt know how to drop the check so i got some help here and with some
new commands as the commande (((((IN))))))

and i created again the check and added licence pro so i can create the module

14. Changez la section de tous les étudiants en "D". Que se passe-t-il? Expliquez

```
mysql> upadte student
-> set section = 'D'
-> where full_name = 'Amine Haddad';
```

i updated the student using same commands i mentioned before

15. Tentez de modifier le specialty_code de "ACAD" en "GEN" dans la table Speciality.
Pourquoi le système refuse-t-il cette modification

```
mysql> update specialty
-> set specialty_code = 'GEN'
-> where specialty_name = 'Informatique G@n@rale';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign
key constraint fails (`usthb_info_db`.`section`, CONSTRAINT
`fk_section_spec` FOREIGN KEY (`specialty_code`) REFERENCES
`specialty` (`specialty_code`))
mysql> // i guess because i did create many tuple with the the
specialty_code =ACAD
-> ^C
mysql> done
-> Terminal close -- exit!
```

i did write the explanation and ill say it again

// i guess because i did create many tuple with the the specialty_code =ACAD